

ELECTRONICS PROJECT

REPORT 2022

By Joshua Anto

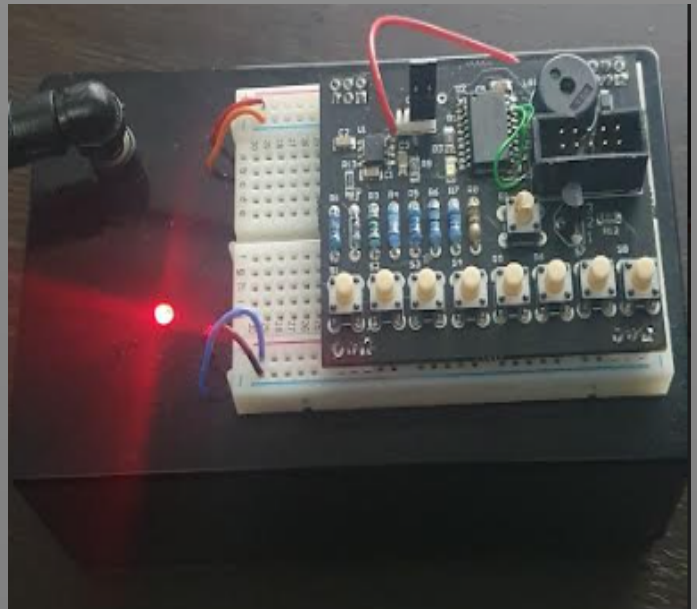
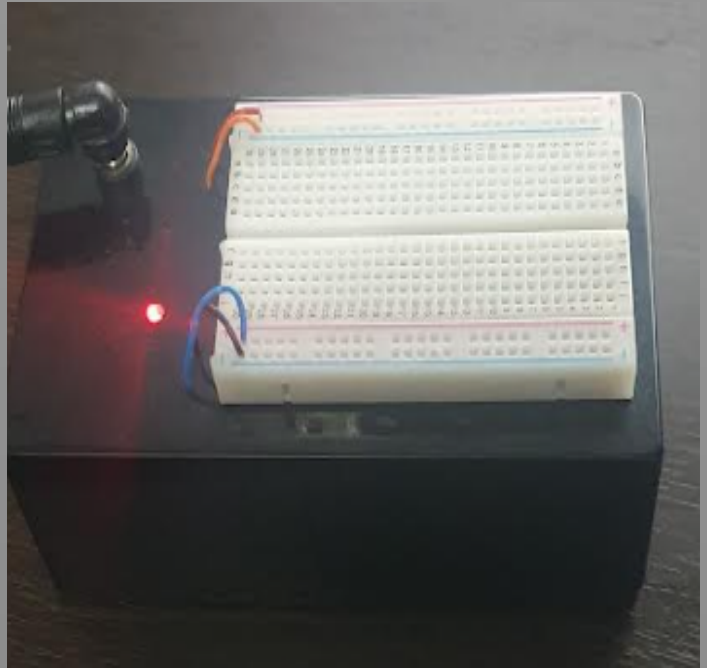


Table of Contents

Table of Figures.....	2
Introduction	2
Half Wave Bridge Dual Rail Power Supply Project.....	3
Theoretical.....	3
Practical.....	4
Simulation	4
Analog and Digital Piano Project	5
Theoretical.....	5
Analogue Piano.....	5
Digital Piano.....	6
Practical.....	6
Simulation	7
Conclusion.....	7
Reference.....	7

Table of Figures

Figure 1: Half-Wave Rectifier Circuit	3
Figure 2: Voltage Regulator	3
Figure 3: NE55 Timer Circuit	5
Figure 4: Chord Frequencies.....	5
Figure 5: Software Button Debouncing	6

Introduction

Project 1 was to design and produce a half-wave-bridge dual rail power supply. The purpose of the Power supply was to create a device that allowed us to draw in AC power from the wall and generate a DC power output, to provide power to components that will be connected to the power supply. This project lay the foundation for powering our Analogue and Digital Piano. The initial process was to convert the sketched schematic into a more detailed schematic in Altium Designer, from there we set up the power and track rules that will provide aid in our PCB design. Altium designer allowed us to design our PCB which was then sent to be manufactured. Once the designs returned, we moved on to placing the components on the board. When all components were placed, the next step was to create the shell that the power supply circuit will sit within. Precise measurements were taken when drilling the holes into the casing, so that all components of the circuit will fit in appropriately. Once the circuit was sitting within the case, the next step was to attach the breadboard and connect the DC output wires to their tracks. This next step ensured our design and circuit was working, we connected the power supply to an AC source and tested using a multimeter whether the voltages produced on the four terminals were as expected. When testing was complete and successful, the next project commenced.

Project 2 was to design and produce an Analogue and Digital piano that works in conjunction with the Power supply. The purpose of the piano was to use two different ways to produce a pulse width modulation and output that to a buzzer to produce a tune. The two different ways being Analogue and Digital. Just like project 1 we used Altium designer to design the schematic with their appropriate rules, which then can be used to design our PCB layout and finally be sent out to the manufacturers to get our design printed. Once printed we assembled the components onto the board making sure to use safety measures to not damage the components or yourself. Once assembled we connected the piano board to the power supply and tested using a multimeter to see if the outputs were as expected. When completed, we tested the analogue push buttons (S1-S8) to make sure they produced a tune on the buzzer. The digital switch (Sw9) was to be coded within Microchip Studio so that every time you would press the switch it will cycle through frequencies which produce a PWM and that will output on the buzzer. However, within this project the analogue and digital methods could not be used simultaneous and required a jumper to choose which mode the user would like.

Half Wave Bridge Dual Rail Power Supply Project

Theoretical

The power supply has a rectifier circuit, which means it can convert the alternating current (AC) into direct current (DC). The rectifier circuit is used to produce +12V which will then flow through a regulator to produce a stable voltage output. Once the AC flows through the rectifier diode, it produces a Half-Wave which then is smoothed out by the filter capacitors and load resistors. A Half-Wave rectifier only allows the current to flow through the diode when the AC voltage is positive. Voltage is positive when the power at T1>T2 within the power supply, this means current will flow from T1 to D1 to produce +12V. When AC power is greater at T1, the current cannot flow through the rectifier diode, D3, as the diode prevents reverse flow. When voltage is negative the AC power at T2>T1, which means current flows through T2 to Ground and from Ground to R3 and then finally through the rectifier diode, D3, this is how -12V is produced. U1 is an integrated circuit that regulates the voltage that was used to power Project 2. The voltage regulator allows us to produce a stable +5V from a +12V input that was produced from the rectifier circuit.

The fuse, F1, is a component that prevents an overload of current through the circuit or short-circuit by melting to cut out the power from the source. This protects the circuit by preventing components from burning out and causing any further damages to the circuit due to excessive current flow. The fuse is connected in series to the AC input, so that the same current flows through all the components.

D4 and D5, are Schottky diodes. Schottky diodes are used to limit the voltage loss, they have very small forward voltage drop between 0.15V and 0.48V. The Schottky diodes protect the circuit by protecting the 5V regulator in case the wrong power was plugged in.

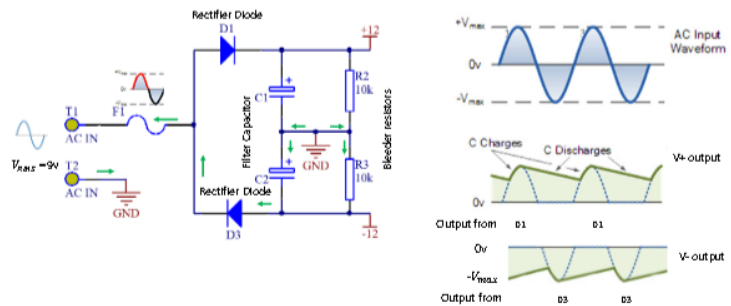


FIGURE 1: HALF-WAVE RECTIFIER CIRCUIT

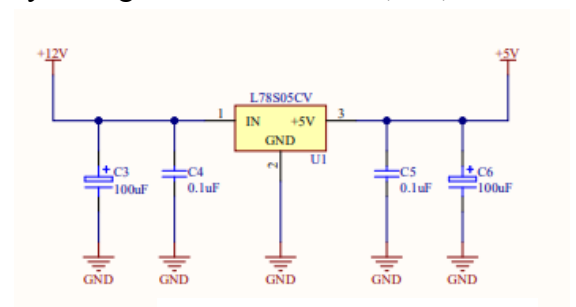


FIGURE 2: VOLTAGE REGULATOR

R4 and R5, are used to protect the +12V and -12V within the rectifier circuit. They protect the circuit by indicating that too much current is flowing through to the circuit by burning out. They act like the fuse, but unlike the fuse they are easier to swap out.

Practical

12V				
Load (mV)	Actual Load (mV)	DC RMS	AC RMS	PK - PK
0	1.85	13.2V	13.0mV	49.6mV
10	10.44	13.0V	15.9mV	68.0mV
20	20.33	12.8V	20.6mV	82.0mV
30	29.73	12.6V	26.2mV	110mV
40	39.64	12.6V	31.5mV	124mV
50	50.36	12.4V	37.1mV	140mV
60	59.94	12.2V	42.0mV	154mV
70	70.9	12.0V	48.1mV	174mV
80	79.5	11.9V	52.6mV	194mV
90	89.2	11.7V	57.3mV	206mV
100	100.7	11.6V	63.2mV	226mV

5V				
Load(mA)	Actual Load(mA)	DC RMS	AC RMS (mV)	Pk to Pk
0	1.55	5.00V	2.61mV	32.0mV
20	20.34	4.80V	1.87mV	26.0mV
40	39.55	4.80V	2.11mV	24.0mV
60	59.5	4.60V	2.81mV	66.0mV
80	80.5	4.77V	2.12mV	38.4mV
100	100.9	4.70V	1.95mV	26.0mV
120	120.6	4.59V	2.98mV	40.0mV
140	140.9	4.60V	2.67mV	43.2mV
160	159.3	4.20V	2.32mV	44.8mV
180	180	3.77V	5.15mV	580mV
200	200.8	4.40V	3.08mV	41.6mV

Simulation

F5 = 698Hz, Resistor value = 70304 Ω

G5 = 784Hz, Resistor value = 47673 Ω

A5 = 880Hz, Resistor value = 27636 Ω

B5 = 988Hz, Resistor value = 975 Ω

C6 = 1047Hz, Resistor value = 154 Ω

The resistors required for these frequencies have a very large resistance or a very small resistance and will be difficult to purchase these components as they are scarce. Therefore, they are not available in terms of stock as other more common resistors are

Digital Piano

For the Digital Piano, it had to be programmed via Microchip studio to be able to produce the tunes and the required frequencies. The IC that was needed to be programmed was the ATtiny 2313A and was required to cycle through different frequencies when S9 was pressed. This was accomplished by using Timer 1, which is a 16-bit Timer/Counter within the IC. For the Timer/Counter a 8MHz clock was used and a prescaler of Clk/8 was set, so that the clock would be able to pulse in 1 sec increments. As the Timer/Counter increments it is constantly checking to see if a compare match has occurred. A compare match occurs when the Timer/Counter = value stored in the OCR1x or ICR1 register. In this case when the Timer/Counter equals the Top value, which was set as ICR1 a rising edge occurs and when the Timer/Counter reaches its Max Value and resets a Falling edge occurs. This combination of rising and falling edges is how a PWM was produced and outputted on the OCR1B line. Every time S9 was a new value was set in the ICR1A and when it matched with the Timer/Counter a PWM was produced on the OCR1B line.

An occurrence within our piano that happened was button bounce. Button bounce is when the electrical contacts in mechanical push buttons often make and break contacts when the button is first pushed and produces a ripple signal. Button debouncing was performed through software measures to remove the ripple signal and produce a clean transition at its output. To

perform Software button debounce a line of code was written to the IC that allowed the MCU to wait for 10ms after it detects the first pulse before checking again for whether the button was pressed.

Another way of Button debouncing is through Hardware, by adding a RC filter circuit.

```
if (Button)
{
    _delay_ms(10); //Debouncing
    if (Button)
    {
        if (count >= 7) //If counter is greater than 7, reset counter and turn off Led
        {
            count = 0;
        }
    }
}
```

FIGURE 5: SOFTWARE BUTTON DEBOUNCING

The S9 oush button is active-low, which means when pressed produces a logical low signal. It uses the pull up resistor within the MCU ATtiny2313A to produce a logical high signal. For pull up resistor to be activated, its corresponding data direction register bit must be set as an input and when the buttons corresponding Pin is pressed it will activate the pull resistor within the MCU to drive a logical high signal.

Practical

Digital (Sw9)	Frequency
	593.5Hz
	667.9Hz
	709.9Hz
	793.7Hz
	895.6Hz
	1001Hz
	1062Hz

Analogue	Frequency
Sw1	421.6Hz
Sw2	479.9HZ
Sw3	559.3Hz
Sw4	599.5Hz
Sw5	685.0Hz
Sw6	753.0Hz
Sw7	859.2Hz
Sw8	880.2Hz

Simulation

Conclusion

Projects 1 and 2 granted insight on the process of designing and building a project based of an individual's electrical knowledge. The process of designing from a schematic and PCB using Altium designer, then building your PCB board with components using soldering techniques. Testing whether our design and assembly were functioning properly using multimeters, played a crucial role in understanding the project process. The steps present in both projects showed us in detail the skills required to be an Engineer

Reference

<https://www.scienceabc.com/innovation/fuse-protect-circuit.html>

<https://byjus.com/physics/schottky-diode/>

<https://circuitdigest.com/article/what-is-bleeder-resistor-and-where-it-is-used>

<https://digilent.com/reference/learn/microprocessor/tutorials/debouncing-via-software/start>